A03 / INDIVIDUAL CONTRIBUTION

Data Engineering & Preprocessing

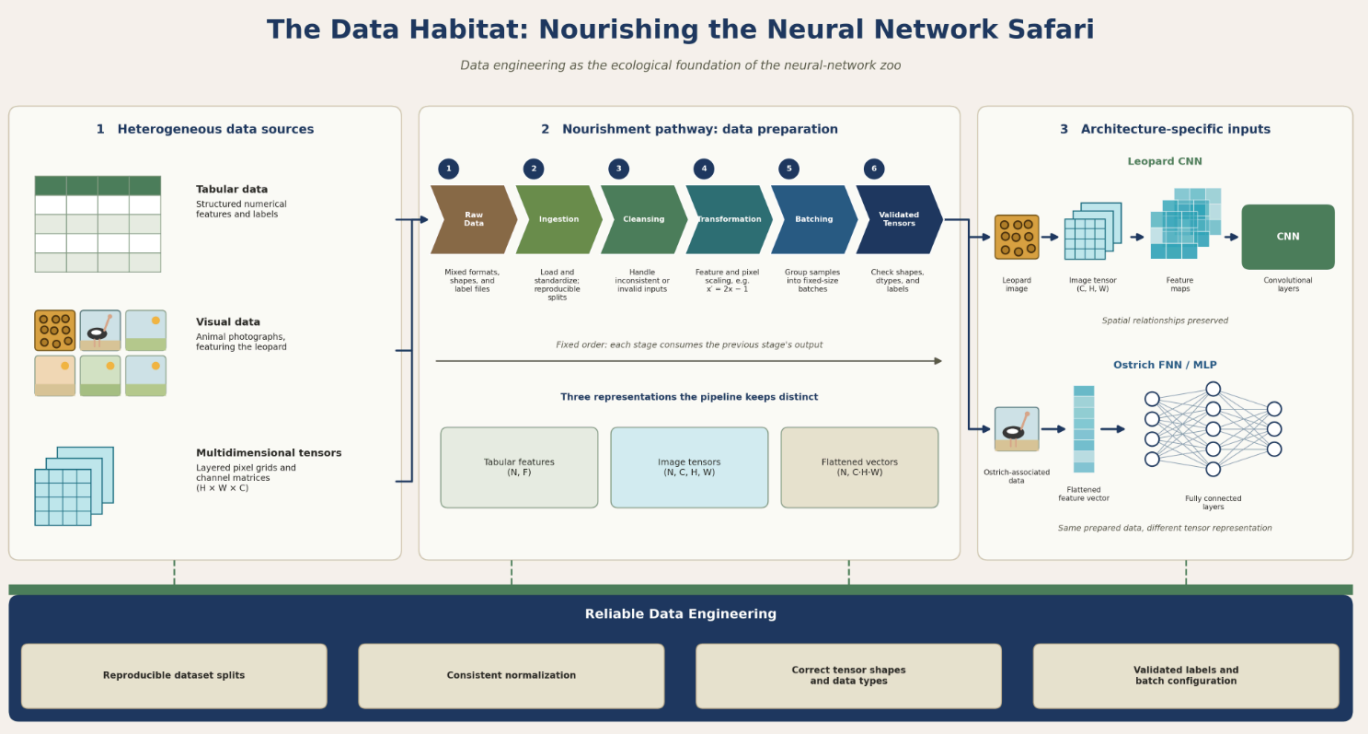
Neural Network Safari

Team Member: Cherluk

Role: Data Engineering & Preprocessing Lead

1. Introduction & Objectives

My primary responsibility for the Neural Network Safari project was establishing a robust data pipeline to feed normalized image and sequence tensors into our modeling framework.



Core Objective

Establish a reproducible, validated data pipeline that delivers correctly formatted tensors to downstream neural network architectures while reducing inconsistencies between preprocessing and model training.

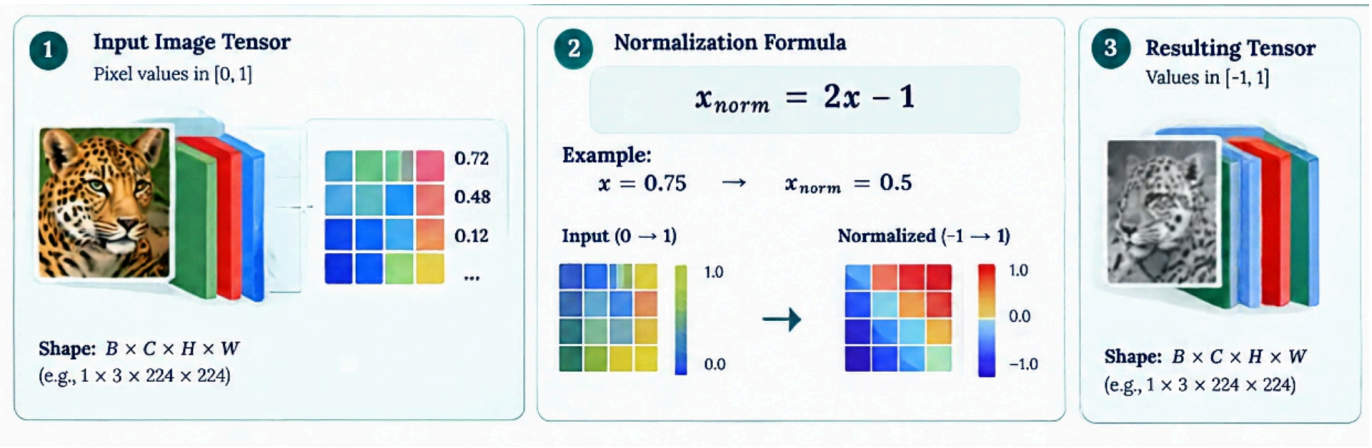
2. Technical Execution & Methodology

To support our architecture zoo, I engineered preprocessing pipelines focused on four core components.

2.1 Dataset Ingestion & Standardization

Handled dataset downloading and loading through PyTorch torchvision, ensuring reproducible training and testing splits and standardizing input workflows for downstream model training.

2.2 Tensor Transformations & Batch Configuration



2.3 Validation Routines

Added validation checks for tensor dimensions, data types, and label mappings before batches were passed to downstream architecture modules, preventing format inconsistencies from interfering with training.

3. Challenges & Solutions

3.1 Challenge: Heterogeneous Input Dimensions

Different architectures require distinct input representations; fully connected neural networks (FNNs) commonly consume flattened feature vectors, whereas convolutional neural networks (CNNs) preserve spatial structure in image tensors.

3.2 Solution: Modular Tensor Transformations

Created modular wrapper functions that reshape tensors according to the target architecture requested by teammates, supporting consistent data handoffs while matching each model's computational design.

4. Reflection & Key Insights

Building this data pipeline highlighted the critical dependency that neural networks have on clean, well-normalized inputs. Even a sophisticated architecture can perform poorly when preprocessing is inconsistent, features are incorrectly scaled, or data-loading bottlenecks prevent efficient training.

Key Insight

Model quality begins before the forward pass. Reliable preprocessing is not simply a preliminary task; it is a foundational part of the learning system.
Correct tensor shapes, consistent normalization, reproducible data splits, and validated labels allow model architectures to learn from meaningful inputs rather than avoidable data inconsistencies.

5. Supporting Evidence & Attribution

Technical Source: zpoo.ipynb, physical Cell 6 (Activity definitions, callback setup attempt, and saved Spider fallback output) and data ingestion modules.

Scope: This report documents the pipeline architecture and data transformations established for the team repository. Individual responsibilities and contributions should be verified against commit histories prior to final submission.